Learning Camera Geometry from Unlabeled Real-World Dynamic Video

Master's Thesis Defense

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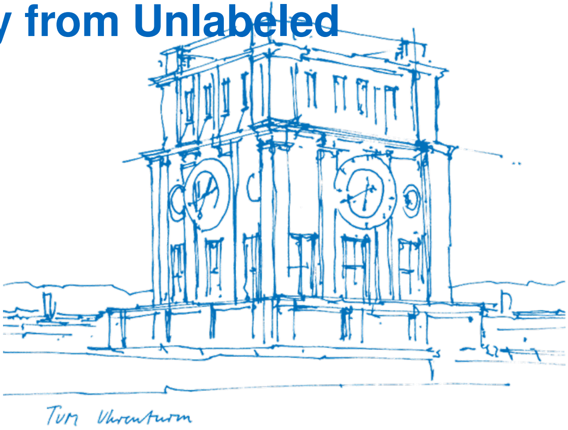

Outline

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Motivation

Recover **camera poses** (R, t) and **calibration** $(\mathbf{K})$ from **unlabeled casual video**

The opportunity

- Billions of casual videos online
- No calibration, no 3D labels
- Self-supervised learning can unlock this data

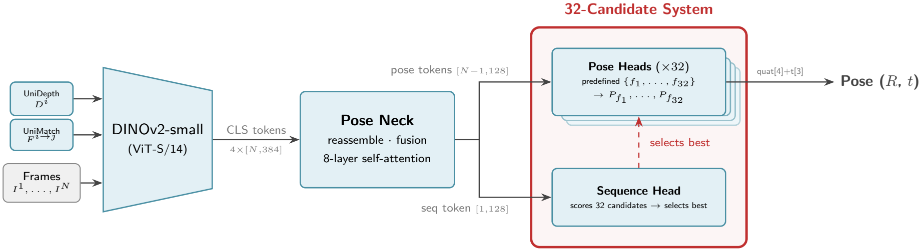
The challenge

- Dynamic objects, unknown cameras
- SfM/SLAM fails on casual video
- Supervised methods don't scale

Strong pretrained models exist for sub-tasks (depth, flow, calibration, pose)

Key idea: Can we **fuse** them into a unified system with multi-frame consistency?

The Calibration Problem in AnyCam



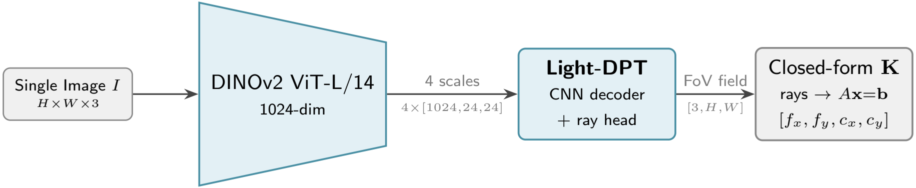
32-candidate focal length system:

- Generates 32 focal length guesses
- Runs flow reprojection for each
- Sequence head selects the best

AnyCam calibration error:

Dataset	Focal length MAPE
Sintel	45.1%
TUM-RGBD	63.7%

AnyCalib — A Path to Better Calibration

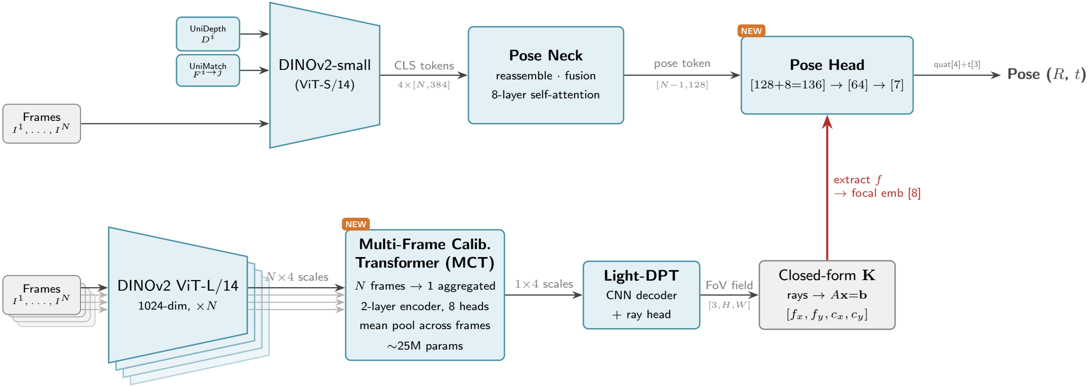


AnyCalib calibration error:

Dataset	AnyCam	AnyCalib
Sintel	45.1%	**30.7%**
TUM-RGBD	63.7%	**11.7%**

- Per-frame only — noisy, inconsistent
- Can we enforce **multi-frame consistency**?
- Can we improve on AnyCalib too?

Our Framework — Fusing Calibration & Pose



Contributions

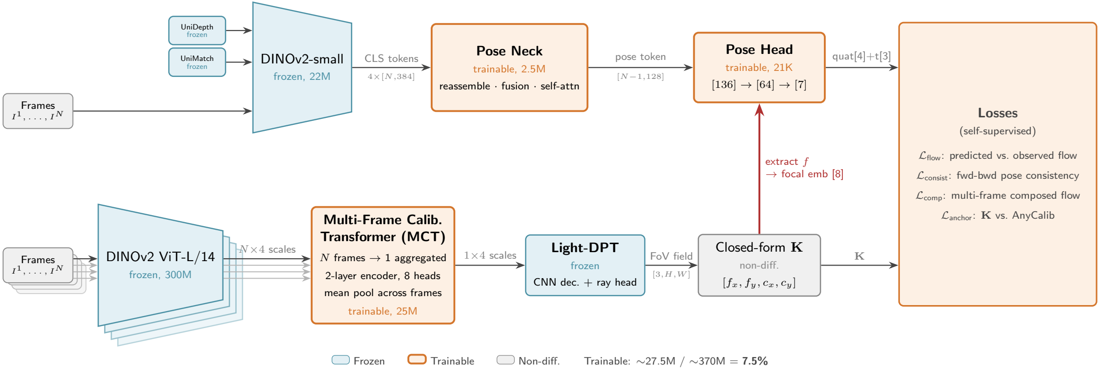
1. Multi-Frame Calibration Transformer (MCT)

- Cross-frame attention on calibration backbone features (4 scales)
- Aggregates N frames $\rightarrow$ 1 consistent calibration
- Architecture-agnostic: inserts between any backbone and decoder
- Only ~ 25M trainable params (7.5% of total)

2. Demonstrated on AnyCam + AnyCalib

- Fuses a self-supervised pose estimator with a calibration network
- Improves *both* pose accuracy and calibration accuracy
- Fully self-supervised — no ground truth labels

Training Setup



Training Losses

Flow Reprojection (self-supervised, main signal):

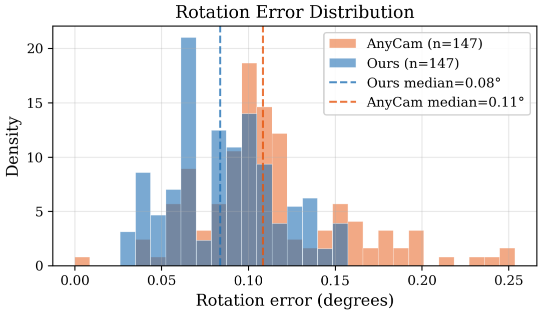
$$\mathbf{x}' = \pi\left(\mathbf{K}\left(R\,\mathbf{K}^{-1}\begin{pmatrix}x\\y\\1\end{pmatrix}d + \mathbf{t}\right)\right), \quad \mathcal{L}_{\text{flow}} = \|\mathbf{x}' - \mathbf{x} - \mathbf{w}_{\text{obs}}\|$$

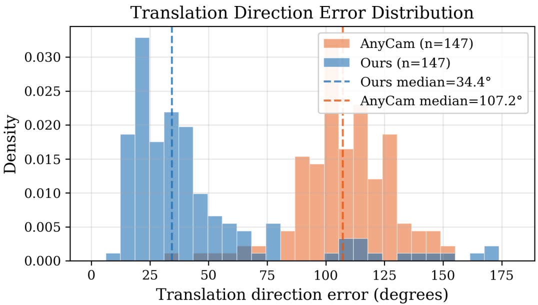
Calibration Anchor — why $\mathbf{K}$ needs constraints:

$$R \approx I,\ \mathbf{t} \approx 0 \quad \Longrightarrow \quad \mathbf{K}^{-1}(d\,\mathbf{K}\mathbf{x} + \mathbf{t}) \approx \mathbf{x} \quad \text{(any } \mathbf{K} \text{ works)}$$

$$\mathcal{L}_{\text{anchor}} = \|\mathbf{K}_{\text{pred}} - \mathbf{K}_{\text{AnyCalib}}\|$$

Results — Pose Error

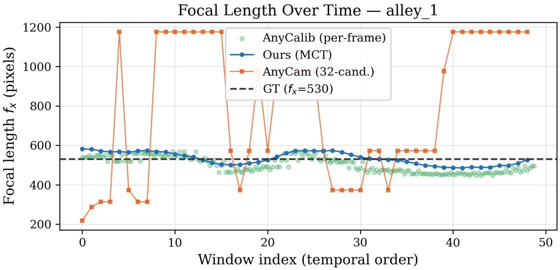


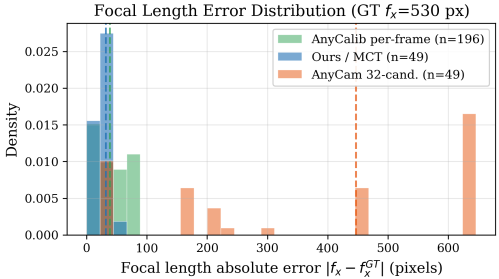


Rotation error	Mean	Median
AnyCam	0.121°	0.108°
Ours	**0.086°**	**0.084°**

Translation error	Mean	Median
AnyCam	108.5°	107.2°
Ours	**44.3°**	**34.4°**

Results — Calibration Error

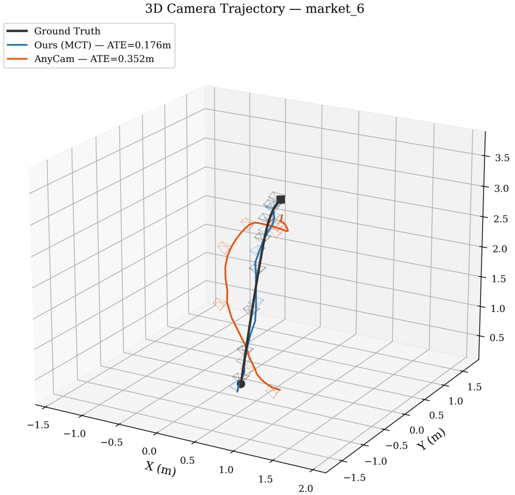




Calibration error	MAE (px)	MAPE
AnyCam	362.1	68.2%
AnyCalib	40.4	7.6%
Ours	**28.5**	**5.4%**

** Sintel `alley_1` sequence*

Trajectory Visualization — Sintel `market_6`



Conclusion & Future Work

Summary:

- Proposed a framework for fusing pretrained calibration & pose models
- MCT: multi-frame calibration consistency via feature-level aggregation
- Demonstrated on AnyCam + AnyCalib: **28% better translation, 16% better calibration**

Future work:

- Longer sequences & non-pinhole camera models
- Joint depth refinement with calibration and pose

Thank you! Questions?